GUIDELINES FOR B. TECH PROJECT-1 REPORT PREPARATION

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान - भोपाल
Maulana Azad National Institute of Technology– Bhopal

GUIDELINES FOR B.TECH PROJECT-1 EXECUTION

Project Timeline

S.No	Activity	Duration
1.	Literature Survey and Topic Finalization	0-2 week
2.	Setting up project objectives and Technologies required	2-4 week
3.	Development Phase-I Implementation of objectives	4-6 week
4.	Development Phase-II Implementation of objectives and project evaluation	6-8 week
5.	Last Date of submission of Progress Report	12 March 2024
6.	Result Analysis and justification of the proposed project	8-10 week
7.	Project Completion and Methodology writeup	10-12 week
8.	Project Report and Synopsis preparation in the prescribed format.	12-14 week
9.	Preparation of final project demonstration	14-16 week

Note: Project meetings will take place once or twice a week. Candidates should consult with their supervisors regarding these meetings.

GUIDELINES FOR B.TECH PROJECT-1 REPORT PREPARATION

Introduction

This document is intended to provide specific and uniform guidelines to the B.Tech students in preparing the project-1 report. The content of the report, is submitted to the institute in partial fulfilment for the award of the degree of Bachelor of Technology. To be acceptable by the institute, it is also imperative that the report meet a uniform format emphasizing readability, concordance with ethical standards and institute-wide homogeneity.

CHAPTER 1
REPORT LAYOUT

The project report has to be organized in the following order.

1. Cover Page
2. Inside Title Page
3. Certificate signed by the Supervisor(s) (in the stipulated format)
4. Declaration signed by the Candidate (in the stipulated format)
5. Acknowledgement
6. Abstract
7. Table of Contents
8. List of Figures
9. List of Tables
10. Abbreviations/ Notations/ Nomenclature (if any)
11. Text of the Report
    Chapter 1
    Chapter 2

…..

…..

12. References
13. Appendices (if any)
14. Non-paper materials (if any)

The formats to be followed for various headings are as follows

1. COVER PAGE: See sample sheet 1. The content, relative font size and locations of various items on the page should match those given in sample sheet 1.

2. INSIDE TITLE PAGE: The inside title page is the same as the cover page except printed on bond paper as given in 2.3.

3. **CERTIFICATE:** See sample sheet 2. The content, relative font size, and locations of various items on the page should match those given in sample sheet 2.

4. **DECLARATION:** See sample sheet 3. The content, relative font size, and locations of various items on the page should match those given in sample sheet 3.

5. **ACKNOWLEDGEMENT:** See sample sheet 4. It should not exceed two pages.

6. **ABSTRACT:** See sample sheet 5.

7. **TABLE OF CONTENTS:** See sample sheet 6.

8. **LIST OF FIGURES:** See sample sheet 7.

9. **LIST OF TABLES:** See sample sheet 8

10. **ABBREVIATIONS/ NOTATIONS/ NOMENCLATURE:** See sample sheet 9.

11. **CHAPTERS:** The chapters may have an Introduction including literature survey, Materials and Methods used, Results, Discussion and Conclusion. See sample sheet 10.

12. **REFERENCES:** To be provided immediately after the last chapter. References should be in APA format (IEEE).

13. **APPENDICES:** See sample sheet 11.

CHAPTER 2
GENERAL GUIDELINES

2.1. Report Size

The report may contain a maximum of about 50 pages, including references and appendices.

2.2. Paper Size

Use A4 size paper (210 mm wide and 297 mm long).

2.3. Paper Quality

White bond paper should be used. Essentially the same quality of paper should be used throughout. Photographs or images with dense colours may be printed on the single side on glossy paper.

2.4. Margins

A margin of 35 mm is to be provided on the left and right sides, whereas top and bottom margins should be 30 mm**.** No print matter should appear in the margin except the page numbers. All page numbers should be centred inside the bottom margin.

2.5. Font

Times New Roman (TNR) 12-point font must be used throughout the running text. Table and figure captions should be 11 points in size, and footnotes should be set at 10 points. Font sizes for various levels of headings are given in section 2.7.

2.6. Line Spacing

The line spacing in the main text should be 1.5. Single line spacing should be given for quotations, abstract, figure captions, table captions, figure legends, footnotes, and references. The equations, tables, figures, and quotations should be set off from the main text both before and after with a spacing of 1.5. Two consecutive paragraphs should be separated by triple line spacing.

2.7. Headings

The following format has to be followed in the heading of chapters and sections.

CHAPTER 3

TITLE PAGE-CENTERED TNR 17-POINT BOLD ALL CAPS

3.1. Section Heading

Left aligned with number, TNR 17 points, bold and leading caps

3.1.1. Second level section heading

Left aligned with number, TNR 14 points, bold and sentence case.

3.1.1.1. Third level section heading

Left aligned with number, TNR 12 points, bold and sentence case.

Fourth-level section heading

Numbered subsections beyond third level are not recommended. However, fourth-level subsection headings may be included without numbering, TNR 12-point font, left aligned and italicized.

Running text should be set in 12-point TNR and fully justified. The first line of the paragraph should have an indentation of 15 mm.

2.8. Table / Figure/equation Format

Tables, figures and equations shall be numbered chapter-wise. For example, the second figure in Chapter 3 will be numbered Figure 3.2. The figure can be cited in the text as Fig. 3.2 or Figure 3.2. However, a consistent citation format should be followed throughout the report. Tables shall be numbered similarly (Table 2 in Chapter 3 will be numbered Table 3.2) and shall be cited in the text as Table 3.2. The figure caption must be located below the figure. The table number and caption must be located above the table. Equations are aligned to the centre of the page with the equation number in the text given at the end of the line within brackets as given below.

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L}\right) \qquad \text{(eq. 2.1)}$$

2.9. Citing References

2.9.1. One author

Monika (2007) developed this method of ……… Subsequently other researchers have adopted this technique (Ramakrishna, 2009; Bhaskar, 2010).

2.9.2. Two authors

Monika and Ram (2008) developed the model of .... Subsequently other researchers adopted this technique (Ramakrishna, 2009; Rao and Ram, 2011)

2.9.3. Multiple authors & more than one publication in a year

"Ram et al. (2005a) has designed the model ….." when given in sentence.

"Model AAB could regulate the control unit more efficiently (Ram et al., 2005b) …." while given in brackets

2.9.4. Citing multiple references

When many authors are cited in sentence it is given as " .... Similar work was also proposed by Singh and Robin (2008); Ram et al. (2009); Prakash (2011)..... "

"Similar work was demonstrated for varied data set by many researchers (Singh and Robin, 2008; Ram et al., 2009; Prakash, 2011)"

2.10. Listing of the References

References are to be listed after the last chapter. They are to be listed in alphabetical order and numbered. Within a reference, the line spacing should be single. Each reference should be separated by one blank line. The reference number should be left aligned. The text of the reference should have an indentation of 10 mm. The reference format to be followed for journal articles, textbooks, conference proceedings, etc., are given below.

2.10.1. Journals

1. Prakas, K. (2011). Feedback and optimal sensitivity: Model reference transformations, multiplicative seminorms, and approximate inverses. *IEEE Transactions on Automatic Control*, 26(2): 301–320.

2. Ram, R., Krishna, S. and Peter, K. (2005a). Risk sensitive estimation and a differential game. *IEEE Transactions on Automatic Control*, 39(9): 1914–1918.

3. Ram, R., Krishna, S. and Peter, K. (2005b). Differential rectification using control points. *IEEE Transactions on Geoscience and Remote sensing*, 55: 914 – 918.

4. Ram, R., Krishnamurthy, P., Prasad, N. and Peter, K. (2009). Risk sensitive estimation model II. *IEEE Transactions on Automatic Control*, 43(15): 355 - 363.

2.10.2. Text books

1. Myers, D. G. (2007). *Psychology* (1st Canadian ed.). Worth: New York.

2. Robin, R. (2008). *Robust Statistics*. Wiley-Interscience: New York.

2.10.3. Conference proceedings

1. Payne, D.B. and Gunhold, H.G. (1986). Digital sundials and broadband technology, In *Proc. IOOC-ECOC,* 1986, pp. 557-998.

2. Singh, K. and Robin, R. (2008). A linear- quadratic game approach to estimation and smoothing. In *American Control Conference,* New York. June 20 – 25, 2008, pp. 2818–2822.

2.10.4. Reports

1. Milton, M and Robert, L. (2004). Atmospheric carbon emission through genetic algorithm, *Environment and Technical Report No.3.*, Indian Meteorological Department., New Delhi.

2.10.5. Online journals with a DOI (Digital Object Identifier)

1. Krebs, D.L. and Denton, K. (2006). Explanatory limitations of cognitive-developmental approaches to morality. *Psychological Review,* 113(3): 672-675. doi: 10.1037/0033-295X.113.3.672

2.10.6. Online journals without a DOI

1. Vicki, G.T., Thomae, M., Cullen, A. and Fernandez, H. (2007). Modeling the hydrological impact on Tropical Forests. *Forest Ecology,* 13(10): 122-132. Retrieved from http://www.uiowa.edu/~grpproc/crisp/crisp.html

2.10.7. Online abstracts

1. Perilloux, C. and Buss, D.M. (2008). Human relationships: Costs experienced and coping strategies deployed. *Evolutionary Psychology,* 6(1): 164-181. Abstract retrieved from http://www.epjournal.net

2.10.8. Online books

1. Perfect, T.J. and Schwartz, B. L. (Eds.) (2002). *Applied metacognition*. Retrieved from http://www.questia.com/read/107598848 (--If DOI is available, use the DOI instead of a URL)

2.10.9. Chapters from a book

1. Krebs, D.L. and Denton, K. (1997). Social illusions and self deception: The evolution of biases in person perception. In J. A. Simpson & D. T. Kenrick (Eds.), *Evolutionary social psychology* (pp.21-48). Hillsdale, NJ: Erlbaum.

2.10.10. Books in print form

1. Snyder, C.R., Higgins, R.L. and Stucky, R.J. (Eds.). (1983). *Excuses: Masquerades in search of grace*. New York, NY: John Wiley & Sons.

2.10.11. Dissertations

1. Mack, S. (2000). “Desperate Optimism” M.S., University of Calgary, Canada.

2.11. Page Numbering

Page numbers for the prefacing materials (inside title page, dedication, certificate, declaration, acknowledgements, abstract, table of contents, etc.) of the report shall be in small Roman numerals and should be centred at the bottom of the pages. The numbering of the prefacing material starts from the inside Title Page. However, the number is not printed on the Inside Title Page. Each new item of the prefacing materials listed above should start on fresh paper on the right page. If the content of the prefacing material exceeds one page, it has to be printed on both sides of the paper by starting from the right side page. For example, if the item “Table of Contents” extends for 5 pages, it should be printed on fresh paper on the right side page with the second page of the “Table of Contents” on the back of the paper and then continued. The page numbers of the prefacing material will be printed in small Roman numerals continuously counting blank pages also. However, the numbers are not printed on the blank pages.

The body of the report, starting from Chapter 1, should be paginated in Arabic numerals and should be centred at the bottom of the pages. The pagination should start with the first page of Chapter 1 and should continue throughout the rest of the report. Each side of a sheet of paper should be counted as a separate page, even if the back side of a sheet of paper is blank. The odd-numbered pages are always on the right and even-numbered pages are always on the left. If the end of a chapter is on an odd page (right side page), the next chapter should start on an odd page, i.e., on a fresh paper, and should be numbered as odd only by counting the blank even page. However, the page number is not printed on the blank pages.

2.12. Printing

Printing of all material in general should be double –sided in black ink with exceptions as indicated in sections 2.3 and 2.11.

2.13. Binding

Project report copies to be submitted for evaluation are to be soft-bounded. The cover page should be printed on glossy white card 300 g/m^2 or above.

2.14. Electronic Copy

An electronic version of the report should be submitted to the Head of the Department and the concerned faculty in charge.

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Maulana Azad National Institute of Technology– Bhopal

Department of Computer Science and Engineering

कंप्यूटर विज्ञान और अभियांत्रिकी विभाग

Session-

"PROJECT-1 REPORT"
On
"Project Title"

Submitted In partial Fulfillment for the degree of Bachelor of Technology in Computer Science and Engineering

SUBMITTED BY:	**GUIDED BY:**
Scholar Name (Scholar No.)	**Supervisor Name**

Sample sheet 2

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान - भोपाल
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DECLARATION

I hereby declare that the work, which is presented in this Project Report, entitled "**Title Name**", in partial fulfilment of the requirements for the award of the degree, submitted in the **Department of Computer Science and Engineering, Maulana Azad National Institute of Technology, Bhopal.** It is an authentic record of my work carried out from __to ___ under the noble guidance of my guide "**Supervisor Name".** The following project and its report, in part or whole, have not been presented or submitted by me for any purpose in any other institute or organization. I hereby declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may possibly occur, I will be the one to take responsibility.

SCHOLAR NAME

SCHOLAR NO

Sample sheet 3

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान - भोपाल
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CERETIFICATE

This is to certify that **“Student Name”,** student of B.Tech 3rd Year (Computer Science & Engineering), have successfully completed their project **"Project Title "** in partial fulfilment of their Bachelor of Technology in Computer Science & Engineering.

Supervisor Name

Sample sheet 4

ACKNOWLEDGEMENT

With due respect, we express our deep sense of gratitude to our respected Guide and project coordinator, **“Supervisor Name”** for his valuable help and guidance.

We are thankful for his encouragement in completing this project successfully. His rigorous evaluation and constructive criticism were very helpful.

We are also grateful to our respected director, **“Director Name”** for permitting us to utilize all the necessary college facilities.

Needless to mention the additional help and support extended by our respected HOD, **“HOD Name”** in allowing us to use the departmental laboratories and other services.

We are also thankful to all the other faculty, staff members, and laboratory attendants of our department for their cooperation and help.

Last but certainly not least, we would like to express our deep appreciation towards our family members and batch mates for providing much-needed support and encouragement.

Place:

Date:

Scholar Name

Scholar No

ABSTRACT

The abstract of the report should be concise, limited to a maximum of 300 words, and presented without any citations, abbreviations, figures, tables, or equations. It should be structured in well-organized paragraphs, according to the author's preference, with each paragraph beginning with a 15 mm indent. There is to be no extra spacing between successive paragraphs. The text must be formatted in Times New Roman font, size 12, and single-spaced.

Additionally, the abstract is required to be provided in both Hindi and English. The Hindi version should precede the English version. This bilingual requirement is in compliance with the directive from the competent authority, as per order no. 2139 dated 08/01/2024.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

ABBREVIATIONS/ NOTATIONS/ NOMENCLATURE

Utmost care should be taken by the project student while using technical abbreviations, notations and nomenclature.

The abbreviations should be listed in alphabetical order as shown below.

AI	Artificial Intelligence
URL	Uniform Resource Locator
LAN	Local Area Network

The meaning of special symbols and notations used in the report should be explained.

$|x|$ - absolute value of x

μ - mean

$\log_n(x)$ - logarithm (x) to the base n

Sample sheet 10

CHAPTER 1

INTRODUCTION

1.1. Time Series Analysis

Time series analysis has become increasingly vital in predictive modeling, given its ability to analyze and forecast based on time-dependent data. The application of time series analysis spans various domains, from economics to meteorology. As Smith and Johnson (2018) highlighted, the accuracy of predictive models significantly improves with the incorporation of time series data, especially in short-term forecasting.

Table 1.1, adapted from Patel and Kumar (2020), presents an overview of common methods used in time series analysis, including ARIMA, Exponential Smoothing, and LSTM networks, along with their respective application areas.

Recent developments in time series analysis emphasize the use of machine learning techniques for enhanced prediction accuracy. For instance, Figure 2.1 illustrates the LSTM network architecture for time series forecasting, as proposed by Zhang and Lee (2019). This method demonstrates substantial improvements over traditional models in handling complex patterns and non-linear relationships in time series data.

Table 1.1 Title of the table (Times New Roman 11)

A[a]	**B**[b]	**C**	**D**

[a] A is admonishment coefficient of total population (Times New Roman 10)

[b] B is Bombardment coefficient of the mean population (Times New Roman 10)

1.1.1 Motivation of the study

The rapid evolution of big data and increasing complexity of global systems have made time series analysis more crucial than ever in various fields. With the advent of real-time data streaming, the ability to analyze and forecast trends efficiently is vital.

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[1] Adapted from Monika and Ram, 2008 (Times New Roman 10)

The satellite image as given in Figure 1.1 shows the area from where samples are collected.

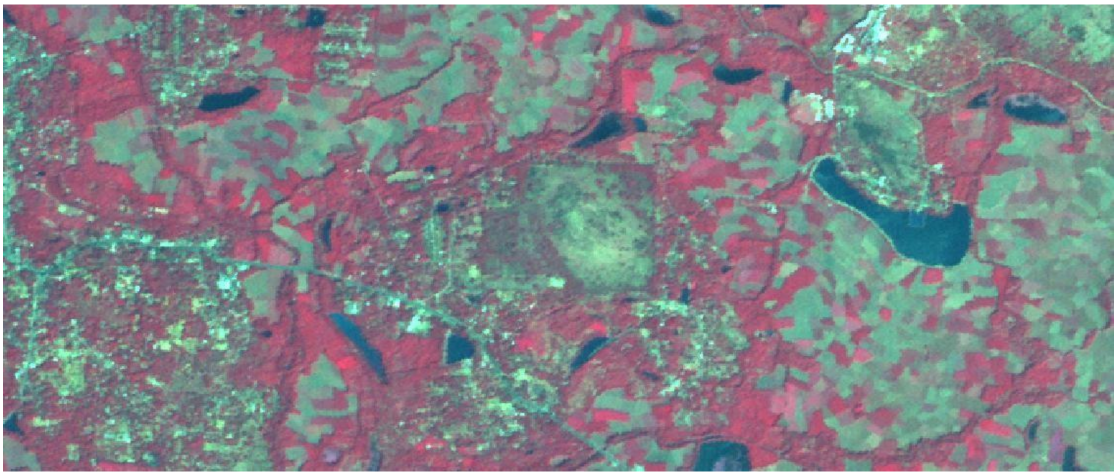

Figure 1.1 Title of the figure (Times New Roman 11)

Sample sheet 11

APPENDIX 1

LIST OF RESPONDENTS TO THE SURVEY

1. IIST
1. NIT
2. JNU
3. MKU
4. KU
5. JNTU
6. Vizag University
7. IIT, Delhi
8. IIT, Mumby
9. IIT Chennai